

ACUGESIC

DESCRIPTION: **ACUGESIC TABLET 50 MG** is a 7 mm square shape, apple green in colour tablet scored with marking 'duo 861' on one side and plain on the other side.
ACUGESIC CAPSULE 50 MG is a black/green capsule with marking 'DUO 861'.

COMPOSITION: **ACUGESIC TABLET 50 MG** : Each tablet contains Tramadol Hydrochloride 50 mg.
ACUGESIC CAPSULE 50 MG: Each capsule contains Tramadol Hydrochloride 50 mg.
 Gelatin for Hard Capsule: Bovine source certified with Malaysia's Halal certification

PHARMACODYNAMICS: The side effects of opioid therapy, manifested as respiratory depression and the development of dependence, which limit the application of potent analgesics, continue to impose on the therapist the difficult task of estimating the benefit-risk relation. Since it has not yet been possible to separate the therapeutically exploitable activity and the undesirable side effects of potent analgesics, it was assumed that the activity and side-effects of potent analgesics were traceable to a common principle of action. Increasing knowledge suggests however that it is possible to distinguish between the therapeutic effects and side effects.

Tramadol (INN) (±)-trans-2-(dimethylaminomethyl)-1-(methoxy-phenyl)-cyclohexanol hydrochloride) is classified as a potent centrally active analgesic. Its structure is related to opiate alkaloids (eg. codeine) and to synthetic opiate analgesics.

Preclinical and clinical studies have shown that the analgesic action derives from the reaction of the drug with the specific opioid receptors of the pain processing system.

Animal experiments led to the assumption that tramadol hydrochloride is a substance in which the therapeutic effects and side effects can to a great extent be differentiated.

In tramadol HCl, with its favourable side effect / analgesic action relation, we now have available for the first time an opioid for the therapy of severe pain with a minimized abuse and dependence potential, and which in therapeutic doses usually induces no respiratory depression. Another advantage of tramadol HCl is its absence of any euphoric component of action – a property which in classical opioids tends to promote abuse and thereby contributes to the development of dependence. Human pharmacological investigations using various methods have shown that tramadol has a distinct analgesic effect. After electrostimulation, 50 mg tramadol proved more effective than 500 mg metamizole or 50 mg flupirtine and, at the same dosage of 75 mg, identical to dextropropoxyphen. The duration of action after dosing with 75 mg tramadol was 3-4 hours. Further human pharmacological studies were performed to establish unwanted effects of tramadol on other organ systems.

Central nervous system: An oral dose of 50 mg tramadol induced a pronounced relaxant – central sedative – effect in electroencephalography (EEG) and electronystagmography (ENG). After administration of 75 mg tramadol, no influence on physical working capacity or psychomotor performance in man was detected.

A double-blind study in which 100 mg tramadol i.m. were administered to 30 test subjects versus placebo, with threefold repetition of the experiment, yielded no evidence for a euphoric effect of tramadol. No significant difference compared to the placebo was determined.

Cardiovascular system: The effects of tramadol on the cardiovascular system were investigated with doses of 50 mg and 100 mg; no major adverse effects on the cardiovascular system were observed. 50 mg tramadol i.v. caused no change in the dp/dt_{max} , and P_{LVED} increased from 8 mmHg to 9 mmHg up to the 40th minute.

Furthermore, a decrease in the pulmonary arterial resistance occurred, so that the influence of tramadol on the circulation was so slight that the authors consider tramadol, with its favourable relation of analgesic action and side effects, to be suitable for pain therapy in acute myocardial infarction.

Tramadol administered preoperatively in an i.v. dose of 0.75 and 1.5 mg / kg body weight in a non-controlled study induced moderate non-significant dose dependent pressure increase in the systemic and pulmonary circulation with a corresponding increase in the peripheral vascular resistance of 25% and a lower increase in the pulmonary arterial resistance.

Five minutes after i.v. injection of 100 mg tramadol, a transient but significant increase in heart rate from a mean 10 min⁻¹, the cardiac index (of 0.4 l/min/m²) and the mean arterial pressure of 4.7 mmHg were observed. The next measurement 10 minutes after the injection showed that the circulatory parameters had already stabilized to the initial level. In a further study, 15 patients received an i.v. dose of 100 mg tramadol and were monitored for 60 minutes for the circulatory effects of tramadol. No circulatory effects could be detected in this study. Twelve patients who received 1.5 mg / kg tramadol i.v. for postoperative pain relief showed virtually no influence on their hemodynamic parameters. Viewed overall, none of the changes was statistically significant. Summing up the effects of tramadol on the greater and lesser circulation and the heart are slight and of such inconsiderable clinical significance that there are no reservations as to its use in painful coronary heart diseases.

Respiratory system: Therapeutic doses of tramadol do not depress respiration. Several studies have shown that tramadol has no significant effect on respiration rate, tidal volume, minute volume, arterial CO₂, ventilatory CO₂ response or mouth occlusion pressure. The influence of potent analgesics on the mouth occlusion pressure and the ventilatory CO₂ response were investigated in 9 test subjects at therapeutic dosages. 50 mg tramadol administered subcutaneously induced no significant deviations in the ventilatory CO₂ response and the mouth occlusion pressure, whereas a subcutaneous dose of 30 mg pentazocine caused a significant decrease in the ventilatory CO₂ response and also depressed the respiratory mechanism.

30 minutes after i.v. administration of 100 mg tramadol to 60 test subjects, no significant change in the stimulation quotients* was determined.

$$*(EO = VE (1 / \text{min})) / P_{ACO_2} \text{ mmHg}$$

100 mg tramadol i.v. administered to 15 patients led to no statistically significant changes in the composition of the blood gases over the 60-minute observation period.

Twelve patients who received an i.v. dose of 1.5 mg tramadol / kg body weight postoperatively for pain relief, showed no statistically significant changes in the arterial blood gases, and no signs of respiratory depression were noted. Following neurolept analgesia and halothane anaesthesia, tramadol was administered by another group of workers to a total of 20 patients in the dose of 1.5 mg / kg body weight. In this non-controlled study, the authors determined a slight, clinically non-relevant decrease in respiratory rate and respiratory amplitude and also an increase in P_{ACO_2} .

Gastrointestinal system: With the aid of endoscopic papillary and biliary duct manometry, the function of the papilla of Vater was characterized before and 10 minutes after i.v. injection of 50 mg tramadol, 30 mg pentazocine, 0.3 mg buprenorphine and placebo (0.9 % NaCl solution). All motility parameters remained unchanged after tramadol dosing, whereas pentazocine induced a statistically significant prolongation of the papillary contraction time and also a statistically significant increase in the papillary rest pressure.

PHARMACOKINETICS: After oral application, tramadol is rapidly and almost completely absorbed. Although a first pass effect is present it is less pronounced than with most other opioid analgesics. These findings explain the good oral efficacy of tramadol.

In the Research Study done by Grunenthal, W. Germany, the originator of Tramadol the absolute bioavailability of Tramadol capsules, Tramadol drops and Tramadol suppositories amounts to 68 ± 13% (capsules and drops) and 78 ± 10% (suppositories). However, as somewhat higher serum concentrations of the pharmacologically active metabolite M₁ are reached after enteral administration than after i.v. application, the analgesic effect after some oral and parenteral administration of the same dose will be similar, despite the reduced serum concentration caused by the first-pass metabolism. In man tramadol is eliminated from the serum renally as well as metabolically; biliary excretion is only a slight significance in young volunteers. Tramadol is metabolized in man and animals mainly by N- and O-demethylation and subsequent conjugation. In contrast to animals much more intact tramadol is excreted with the urine in man; the cumulative renal excretion of tramadol is about 10% (p.o.) and 15% (i.v.). With the exception of the metabolite O-demethyl-tramadol (M₁), all metabolites are pharmacologically inactive.

Tramadol is rapidly distributed, giving a rapid onset of effect after intravenous administration. The biological half-life of the terminal phase in man (5.5 h) is within therapeutically favourable range. The main metabolites of tramadol are eliminated from the serum with comparable half-lives. Tramadol and its metabolites are excreted mainly entirely via the kidneys. On multiple administration of tramadol at six-hour intervals, the steady-state serum concentrations are rapidly reached. The steady-state C_{max} value is about twice as high as after single administration. There is no danger of undesirable accumulation of tramadol even with somewhat shorter intervals of administration. A significant increase in the incidence of side-effects due to the accumulation of metabolites on repeated administration is unlikely since the main tramadol metabolites are eliminated in the terminal phase at the same rate as the parent drug. The binding of tramadol to the proteins of human serum is not very marked. The fraction of protein-bound tramadol only amounts to about 4%. Thus, it does not exercise any major influence on the pharmacokinetics of the substance and can be disregarded with respect to drug interactions.

INDICATIONS: Severe acute and chronic pain.

RECOMMENDED DOSAGE: The dosage should be adjusted to the intensity of the pain. Unless otherwise prescribed, TRAMADOL should be taken as follows - independent of meals:

Adults and adolescents (12 years and older): Acugesic is not approved for use in patients below 12 years old.

Paediatric population: The safety and efficacy of Acugesic has not been studied in the paediatric population. Therefore, use of Acugesic is not recommended in patients under 12 years of age.

Single dose for adults and adolescents over 14 years of age: 1 capsule/tablet to be taken with a little liquid. This is usually sufficient to relieve the pain. If, however, pain relief is unsatisfactory, a further TRAMADOL capsule/tablet can be taken after about 30-60 minutes. In general, the daily dose should not exceed 400 mg Tramadol HCl (equivalent to 8 TRAMADOL capsules/tablets). In impaired renal or hepatic function it may be necessary to adjust the dose.

Duration of treatment: During long-term treatment with TRAMADOL the possibility of dependence cannot be entirely excluded. Therefore, the physician is to decide on the duration of treatment and whether the preparation is to be withdrawn temporarily. TRAMADOL should not be given for longer than therapeutically necessary.

ROUTE OF ADMINISTRATION: ORAL

CONTRAINDICATIONS: Acute intoxication with alcohol, hypnotics, analgesics or other CNS-acting drugs and hypersensitivity to tramadol or any excipients.

Note: In accordance with currently prevailing recommendation medication with the preparation during pregnancy should only be resorted to after careful consideration of the risks. So far no reports are available on its use during lactation. The preparation should be used with care in patients with increased reactivity to opioids.

- Children younger than 18 years to treat pain after surgery to remove the tonsils and/or adenoids.

- Adolescents between 12 and 18 years who are obese or have conditions such as obstructive sleep apnea or severe lung disease, which may increase the risk of serious breathing problems.

WARNING & PRECAUTIONS: Even when administered according to instruction the preparations may affect the reaction ability of the patient to such an extent that his capacity to drive or operate machines may be impaired. This applies particularly in conjunction with alcohol. Should be used with caution in patients with myasthenia gravis and sensitivity to opiates tramadol hydrochloride..

Caution with patients with severe renal and hepatic impairment, head injury, increased intracranial pressure, and shock.

Children: Under 12 is not recommended.

Paediatric population: The safety and efficacy of Acugesic has not been studied in the paediatric population. Therefore, use of Acugesic is not recommended in patients under 12 years of age.

Respiratory depression: Administer Acugesic cautiously in patients at risk for respiratory depression, including patients with substantially decreased respiratory reserve, hypoxia, hypercapnia, or pre-existing respiratory depression, as in these patients, even therapeutic doses of Acugesic may decrease respiratory drive to the point of apnea. In these patients, alternative non-opioid analgesics should be considered. When large doses of tramadol are administered with anaesthetic, medications or alcohol, respiratory depression may result. Respiratory depression should be treated as an overdose. If naloxone is to be administered, use cautiously because it may precipitate seizures.

Cytochromes P450 (CYP) 2D6 Ultra-Rapid Metabolism: Some individuals may be CYP2D6 ultra-rapid metabolisers. These individuals convert tramadol more rapidly than other people into its more potent opioid metabolites O-desmethyltramadol (M1). This rapid conversion could result in higher than expected opioid-like side effects including life-threatening respiratory depression. The prevalence of this CYP2D6 phenotype varies widely and has been estimated at 0.5 to 1% in Chinese, Japanese and Hispanics, 1 to 10% in Caucasians, 3% in African Americans, and 16 – 28% in North Africans, Ethiopians, and Arabs. Data are not available for other ethnic groups.

Risk from Concomitant Use with Benzodiazepines

Profound sedation, respiratory depression, coma, and death may result from the concomitant use of Acugesic with benzodiazepines. Observational studies have demonstrated that concomitant use of opioids and benzodiazepines increases the risk of drug-related mortality compared to use of opioids alone. Because of these risks, reserve concomitant prescribing of these drugs for use in patients for whom alternative treatment options are inadequate.

If the decision is made to newly prescribe a benzodiazepine and an opioid together, prescribe the lowest effective dosages and minimum durations of concomitant use.

If the decision is made to prescribe a benzodiazepine in a patient already receiving an opioid, prescribe a lower initial dose of the benzodiazepine than indicated in the absence of opioid, and titrate based on clinical response.

If the decision is made to prescribe an opioid in a patient already taking a benzodiazepine, prescribe a lower initial dose of the opioid, and titrate based on clinical response.

Follow patients closely for signs and symptoms of respiratory depression and sedation. Advise both patients and caregivers about the risks of respiratory depression and sedation when Acugesic is used with benzodiazepines. Advise patients not to drive or operate heavy machinery until the effects of concomitant use of the benzodiazepine have been determined. Screen patients for risk if substance use disorders, including opioid abuse and misuse, and warn them of the risk for overdose and death associated with the use of benzodiazepines (See Drug Interactions).

Serotonin Syndrome with Concomitant Use of Serotonergic Drugs

Cases of serotonin syndrome, a potentially life-threatening condition, have been reported during concurrent use of Acugesic with serotonergic drugs (See Interactions with Other Medicaments). This may occur within the recommended dosage range.

Serotonin syndrome symptoms may include mental-status changes (e.g. agitation, hallucinations, coma), autonomic instability (e.g. tachycardia, labile blood pressure, hyperthermia), neuromuscular aberrations (e.g. hyperreflexia, incoordination) and/or gastrointestinal symptoms (e.g. nausea, vomiting, diarrhoea) and can be fatal (See Interactions with Other Medicaments). The onset of symptoms generally occurs within several hours to a few days of concomitant use, but may occur later than that. Discontinue Acugesic if serotonin syndrome is suspected.

Adrenal Insufficiency

Cases of adrenal insufficiency have been reported with opioid use, more often following greater than one month of use. Presentation of adrenal insufficiency may include non-specific symptoms and signs including nausea, vomiting, decreased appetite, fatigue, weakness, dizziness, and low blood pressure. If adrenal insufficiency is suspected, confirm the diagnosis with diagnostic testing as soon as possible. If adrenal insufficiency is diagnosed, treat with physiologic replacement dosing of corticosteroids. Wean the patient off of the opioid to allow adrenal function to recover and continue corticosteroid treatment until adrenal function recovers. Other opioids may be tried as some cases reported use of a different opioid without recurrence of adrenal insufficiency. The information available does not identify any particular opioids as being more likely to be associated with adrenal insufficiency.

Sexual Function/Reproduction

Long term use of opioids may be associated with decreased sex hormone levels and symptoms such as low libido, erectile dysfunction, or infertility (See Postmarketing Experience)

INTERACTION WITH OTHER MEDICAMENTS: On the concomitant administration of TRAMADOL with substances which also act on the central nervous system (eg. tranquillizers, hypnotics) the sedative effects (fatigue) may be intensified. At the same time, however, combining TRAMADOL with tranquillizer, for example will probably have a favourable effects on pain sensation. TRAMADOL should not be used in patients receiving MAO inhibitors.

Information for patient: TRAMADOL is a potent drug for relief of pain, eg. in wound pain, fractures, severe nerve pain, tumor pain, heart attack. It should not be used for minor pain. The effect sets in quickly and lasts for some hours.

Benzodiazepines

Due to additive pharmacologic effect, the concomitant use of opioids with benzodiazepines increases the risk of respiratory depression, profound sedation, coma and death.

The concomitant use of opioids and benzodiazepines increases the risk of respiratory depression because of actions at different receptor sites in the central nervous system that control respiration. Opioids interact primarily at μ -receptors, and benzodiazepines interact at GABA_A sites. When opioids and benzodiazepines are combined, the potential for benzodiazepines to significantly worsen opioid-related respiratory depression exists.

Reserve concomitant prescribing of these drugs for use in patients for whom alternative treatment options are inadequate (see Warnings and Precautions).

Limit dosage and duration of concomitant use of benzodiazepines and opioids, and follow patients closely for respiratory depression and sedation.

Serotonergic Drugs

The concomitant use of opioids with other drugs that affect the serotonergic neurotransmitter system has resulted in serotonin syndrome. If concomitant use is warranted, carefully observe the patient, particularly during treatment initiation and dose adjustment. Discontinue Acugesic if serotonin syndrome is suspected. Examples of serotonergic drugs are

selective serotonin reuptake inhibitors (SSRIs), serotonin and norepinephrine reuptake inhibitors (SNRIs), tricyclic antidepressants (TCAs), triptans, 5-HT₃ receptor antagonists, drugs that affect the serotonin neurotransmitter system (e.g. mirtazapine, trazodone, tramadol), monoamine oxidase (MAO) inhibitors (those intended to treat psychiatric disorders and also others, such as linezolid and intravenous methylene blue) (See Warnings and Precautions).

USE IN PREGNANCY & LACTATION:

Use in Pregnancy: Tramadol has been shown to cross the placenta. There are no adequate and well-controlled studies in pregnant women. Safe use in pregnancy has not been established. Acugesic is not recommended for pregnant women.

Use in Lactation: Approximately 0.1% of the maternal dose of tramadol is excreted in breast milk. In the immediate post-partum period, for maternal oral daily dosage up to 400 mg, this corresponds to a mean amount of tramadol ingested by breast-fed infants of 3% of the maternal weight-adjusted dosage. For this reason tramadol should not be used during lactation or alternatively, breast-feeding should be discontinued during treatment with tramadol. Discontinuation of breast-feeding is generally not necessary following a single dose of tramadol.

TRAMADOL is not recommended for obstetric preoperative or for post-delivery analgesia in nursing mothers, because its safety in infants and newborns has not been studied. Low level of TRAMADOL has been detected in breast milk.

SIDE EFFECTS: Sweating, dizziness, nausea, vomiting, dry mouth, and fatigue may occur after TRAMADOL administration. It is also possible that the preparation may affect the cardiovascular system in rare cases. Undesirable effects occur particularly when the patient is under physical strain.

Special note: Respiratory depression has so far not been observed during treatment with TRAMADOL. However, it cannot definitely be ruled out if the recommended dosage is considerably exceeded or on the concomitant administration of other centrally depressant drugs.

Respiratory depression: Rare

Postmarketing Experience:

Serotonin syndrome (See Warnings and Precautions)

Adrenal insufficiency (See Warnings and Precautions)

Androgen deficiency: Cases of androgen deficiency have occurred with chronic use of opioids. Chronic use of opioids may influence the hypothalamic-pituitary-gonadal axis, leading to androgen deficiency that may manifest as low libido, impotence, erectile dysfunction, amenorrhea, or infertility. The causal role of opioids in the clinical syndrome of hypogonadism is unknown because the various medical, physical, lifestyle, and psychological stressors that may influence gonadal hormone levels have not been adequately controlled for in studies conducted to date. Patients presenting with symptoms of androgen deficiency should undergo laboratory evaluation.

Infertility: Chronic use of opioids may cause reduced fertility in females and males of reproductive potential. It is not known whether these effects on fertility are reversible.

SYMPTOMS AND TREATMENT OF OVERDOSE: So far one case of an accidental administration of tramadol hydrochloride in an excessively large dose is known. After accidental subcutaneous administration of a total of 300 mg of the active constituent followed shortly afterwards by an oral dose of 300 mg restlessness, anxiety, tremor, and increased reflexes occurred. These manifestations were alleviated by treatment with pentobarbitone. Blood pressure and pulse rate had not been influenced. Following oral administration of therapeutic doses of tramadol hydrochloride in tablets, fatigue, nausea, vomiting, vertigo, headache, and dryness of the mouth were occasionally observed.

Therapy: Capacity of dialysis and behaviour on forced diuresis: so far not determined.

Symptomatic measures (in convulsion diazepam = Valium 10 – 20 mg is administered intravenously or intramuscularly); as an antidote levallorphan = Lorfan or naloxone = Narcantil.

STORAGE CONDITIONS:

Acugesic Tablet 50 mg: Store below 30°C. Protect from light. KEEP OUT OF REACH OF CHILDREN. (JAUHKAN DARI KANAK-KANAK).

Acugesic Capsule 50 mg: Store below 30°C. Protect from light. KEEP OUT OF REACH OF CHILDREN. (JAUHKAN DARI KANAK-KANAK).

SHELF LIFE: Please refer to outer package

PACK SIZE:

Pack in blister of 10x10's

PRODUCT REGISTRATION HOLDER & MANUFACTURER:

DUOPHARMA (M) SDN BHD

Lot 2599 Jalan Seruling 59, Kawasan 3, Taman Klang Jaya,
41200 Klang, Selangor, Malaysia.